

20 In which of the following cases does an electric field do positive work on a charged particle?

- A** A positive charge moves to a point of higher electric potential.
- B** A negative charge moves opposite to the direction of the electric field.
- C** A positive charge completes one circular path around a stationary positive charge.
- D** A negative charge moves perpendicular to the electric field between two parallel and oppositely charged plates.

Ans: B

Option B is the only one where the E-field does positive work. In option A and D, it is the external agent that does positive work. For option C, since force and displacement is perpendicular, no work is done.

